

AI Security Lab — Level 04: Give It a Role

Vulnerability Overview

- **Vulnerability Type:** Persona Adoption / Context Manipulation
- **Objective:** Assign alternative personas or roles to the AI model and evaluate whether contextual role instructions can influence its behavior or cause deviations from intended constraints.

Technical Analysis

LLMs generate responses based on the complete conversational context available to them. User-supplied role or persona instructions can therefore influence the model's behavior, particularly when the application's security architecture relies primarily on natural-language instructions to enforce behavioral restrictions.

An attacker may attempt to establish an alternative persona or contextual framing designed to influence how the model interprets subsequent instructions.

However, persona adoption should not be treated as a guaranteed security bypass. The effectiveness of such techniques depends on the model, system instructions, guardrails, application architecture, and surrounding security controls.

Security Impact

Potential impacts include:

- Behavioral deviation from intended application policies.
- Increased susceptibility to prompt-injection attacks.
- Inconsistent enforcement of system-level restrictions.
- Potential disclosure of information or functionality that should remain restricted.

Remediation

- Implement strong **system-level guardrails**.
- Enforce authorization through the application and API layer rather than relying exclusively on natural-language instructions.
- Maintain clear trust boundaries between system instructions and user-controlled content.
- Use deterministic security controls for privileged operations.
- Conduct adversarial testing against persona-based and contextual manipulation techniques.

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